

File Normalisation (CPP-026)

CPP-Identifier	CPP-026
CPP-Label	File Normalisation
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Change history	Comments
Version 1.0.0 - 2025-08-29	Milestone version
Version 1.1.0 - 2026-03-30	Migration to XML
Version 1.2.0 - 2026-04-07	Version updated

1. Description of the CPP

The TDA performs operations on the data Objects prior to ingest in order to comply with its format requirements.

Inputs and outputs

Input(s)	
Data	Submission Information Package
	File
Metadata	Technical Metadata (Format identifier)
Documentation/ guidance	File format policy - Preferred formats
Output(s)	
Data	New File
	Updated Submission Information Package
Metadata	Provenance metadata (Normalisation events, original and normalised formats, timestamp)

Definition and scope

In the context of digital preservation, File Normalisation refers to the process of converting incoming *Files* into standardised, well-supported formats that are more likely to be usable and accessible in the long term. These formats are captured in the preferred preservation formats policy of the TDA. The goal is to ensure the long-term usability, accessibility, and intelligibility of digital content by avoiding dependence on obsolete or proprietary formats and start the repository with a consistent, manageable and sustainable set of formats from the start.

File Normalisation can be driven by an identified risk, in which case it requires that some formats have been assessed a format risk. For each of these formats at risk an alternative format should be available as well as the necessary tools to convert *Files* from the format at risk to its alternative format.

File normalisation policy can also be driven by specific organisational requirements like the availability of specialised applications, viewer preferences (e.g. JPEG2000 for IIIF), or the ability to make subsequent processes easier and more efficient such as creating derivatives, tracking and managing risks and obsolescence, and monitoring and reporting of formats in the archive.

If *Files* are submitted in a format not part of the list of preferred preservation formats and there is no normalisation path available to convert them to a supported format, the outcome is undefined. It is up to the TDA to decide what happens.

It may be that the original *File* is submitted in the *SIP* along with the normalised *File*. In this case, the original *File* can only be in bit-level preservation. Reasons for this may be, for

example, to ensure the authenticity of the digital *Object*, because it is anticipated that the original *File* format is expected to be useful for the designated community for a reasonable time.

Normalisation differs from other processes that create new *Files* or *Representations*:

- Unlike Normalisation, **Creation of Derivatives** reproduces only the information and significant properties of the original that are useful to address specific needs of the designated community. Thus, the output of **Creation of Derivatives** may lack part of the information that is not required to satisfy said needs.
- Whereas Normalisation aims at reducing the number of formats by converting *Files* or *Representations* to preservation formats defined by the institutional formats policy, **File Migration** is performed to address a specific preservation risk.

Process description

Trigger event(s)

Trigger Event	CPP-identifier
Submission of an <i>Information Package</i>	CPP-029 (Ingest)
Change in file format policy - Preferred formats	CPP-018 (Community Watch)

Step-by-step description

No	Supplier	Input	Steps	Output	Customer
↓ sequence					
1	↔ alternative - Alternative triggers				
	1.a	CPP-029 (Ingest)	Each <i>File</i> in the <i>Information Package</i>	Perform the Format Identification process on the <i>File(s)</i>	
	1.b	CPP-018 (Community Watch)	Change in file format policy - Preferred formats	Find <i>Files</i> that require change of normalisation by format, then process each one	
2	CPP-008 (File Format Identification)	Detected format(s)	Compare the detected format with the list of preferred formats	Complete match: Process completed (step 6)	
		File format policy - Preferred formats		Deviation: Determine if the policy provides a normalisation plan (i.e. sequence of tools and configuration parameters to use on the <i>File</i>) (step 3)	

No	Supplier	Input	Steps	Output	Customer
3	CPP-012 (Risk Mitigation)	Normalisation path	Execute the normalisation plan	Normalised <i>File</i>	
4		Validation plan	If present, run the validation plan to determine if the normalised <i>File</i> provides a similar rendition like the original <i>File</i> . If no validation plan is available, it is up to the TDA's policy to decide if manual inspection is needed.	Validation result: OK (step 5a) Validation result: Error (5b)	
5	↩ <i>alternative</i> - Alternative validation results				
5.a		OK	Replace the original <i>File</i> with the normalised <i>File</i> (or add it) in the <i>SIP</i> or <i>AIP</i> and adapt the <i>Fixity metadata</i> and <i>Technical metadata</i>	updated <i>SIP</i> or <i>AIP</i>	
5.b		Error	Report the error for the system or operator to decide further action	Stop the process	
6			Log the normalisation event and its outcome	<i>Provenance metadata</i> in the <i>SIP</i>	

Rationale(s) and worst case(s)

Rationale	Impact of inaction or failure of the process
Store information in supported formats	Format migration at a later point could be impossible
Anticipate on formats becoming less popular	Attempt to migrate data when the tools to do so are no longer available or knowledge about the format is lost
Reducing complexity and risk by normalising to a smaller set of well-chosen formats, TDAs significantly reduce the technical overhead and resources needed for preservation activities.	TDA must maintain too many file formats, which each in turn needs specific knowledge, tools and ongoing maintenance to ensure continued access

2. Dependencies and relationships with other CPPs

Dependencies

CPP-ID	CPP-Title	Relationship description
CPP-008	File Format Identification	The format information of the <i>File</i> in question is one of the deciding input parameters when considering File normalisation.
CPP-012	Risk Mitigation	CPP-012 is in charge of designing normalisation paths whose output would retain all significant properties.
CPP-022	Significant Properties Definition	Like Format migration, File Normalisation should be evaluated based on significant properties.
CPP-023	Risk Properties Definition and Extraction	The level of risk of the detected format is one of the factors that drives the preferred preservation formats policy.

Other relations

Relation	CPP-ID	CPP-Title	Relationship description
May require	CPP-005	Identifier Management	A normalised file format may be assigned with a new PID.
May be required by	CPP-029	Ingest	The ingest may require that the digital <i>Objects</i> are first normalised before ingestion.
Facilitates	CPP-015	Emulation and Rendering Tools	Restricting the formats to those that are well supported by the tools improves the success of this process.
Affinity with	CPP-014	File Migration	Normalisation is performed at Ingest and aims at reducing the number of formats preserved by converting <i>Files</i> or <i>Representations</i> to preservation formats defined by the institutional formats policy, while Format migration is performed after ingest to address a specific preservation risk.
Not to be confused with	CPP-028	Creation of Derivatives	Unlike Normalisation, Creation of Derivatives reproduces only the information and significant properties of the original that is useful to address specific needs of the Designated Community. Thus, the output of Creation of Derivatives may lack part of the information that is not required to satisfy said needs. Normalisation aims to maintain all the significant properties of the original.

3. Links to frameworks

Certification

Certification framework	Term used in framework to refer to the CPP	Section
CTS	normalising	page 7 of 19 - Levels of Curation: it regards normalisation a requirement to achieve Curation Levels above B.
Nestor Seal	normalisation	page 34: normalisation mentioned as required to document for c17 Authenticity: Ingest
ISO 16363	normalisation mentioned in examples	4.1.1 - preservation objectives 4.2.2 - describing transformations 4.2.3.3 - provenance information

Other frameworks and reference documents

Reference Document	Term used in framework to refer to the process	Section
OAIS	file format conversion	OAIS mentions in the Generate AIP function of the Ingest Functional Entity: 'may involve file format conversions' (page 4-7)
PREMIS	Normalization	Glossary, page 271: Normalization: Form of Migration in which a version of a Digital Object is created in a new Format with properties more conducive to preservation treatment. Normalization is often implemented as part of the Ingest process.

4. Reference implementations

Publicly available documentation

Institution	Organisation type	Language	Hyperlink
CSC - IT Center for Science Ltd., FI	Non-commercial digital preservation service	English	https://urn.fi/urn:nbn:fi-fe2025040925236 (section 7.7)
Archivematica, CA	Digital preservation system	English	https://www.archivematica.org/en/docs/archivematica-1.17/user-manual/ingest/ingest/#normalize
		English	https://www.archivematica.org/en/docs/archivematica-1.17/user-manual/preservation/preservation-planning/#normalization (Tableau1_3)